SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE
SUBJECT : Data Visualization with Power BI and Tableau

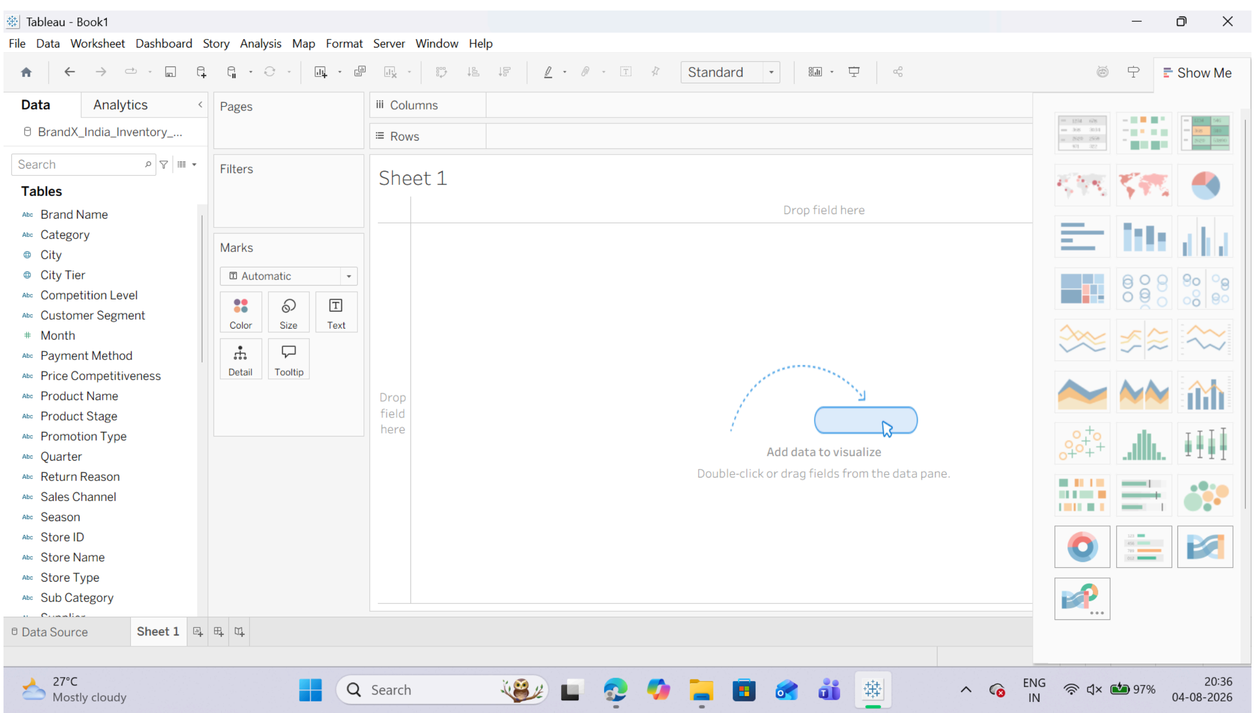
PRACTICAL NO 4

AIM : Performing Data Analysis using Calculations in Tableau

a) Create basic calculated fields using formulas
b) Perform row-level calculations
c) Perform aggregate-level calculations
d) Implement FIXED Level of Detail (LOD) calculations
e) Use INCLUDE and EXCLUDE LOD expressions
f) Create and use parameters for dynamic analysis
g) Apply calculations to fix data issues and enhance insights

A: Creating Basic Calculated Fields Using Formulas

1. Connecting to the Dataset
2. Click on Sheet 1 at the bottom-left corner to enter your workspace.



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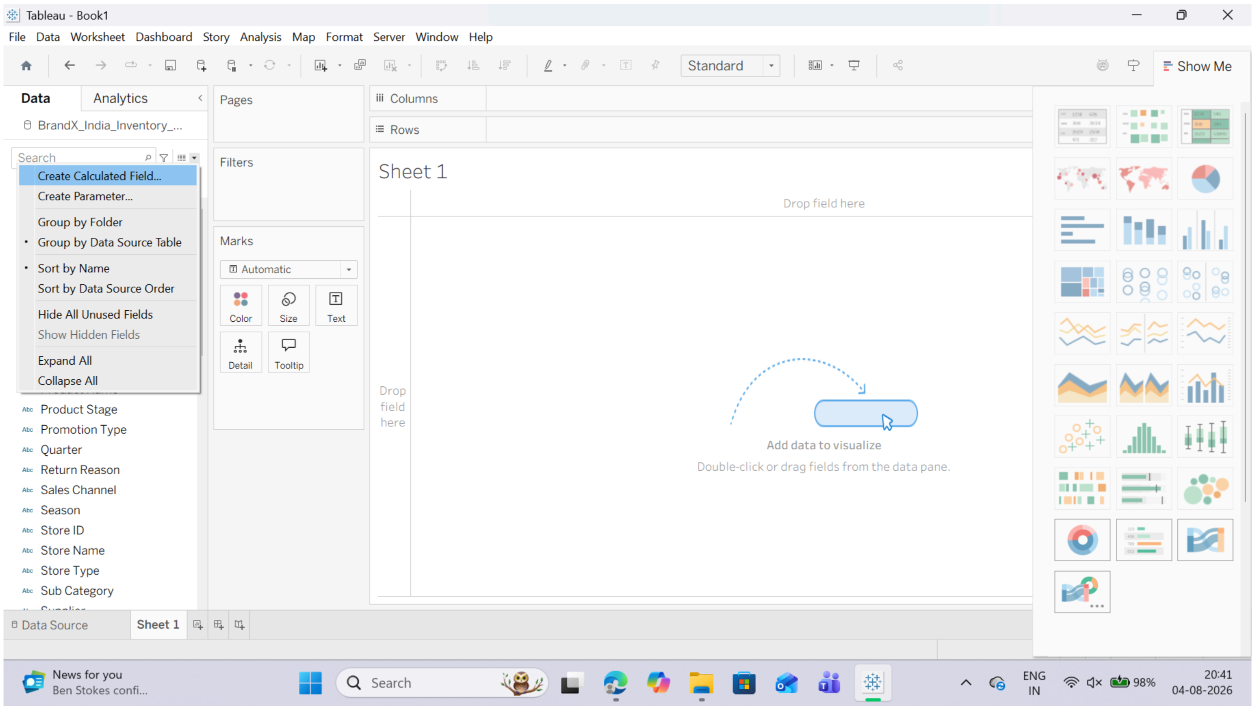
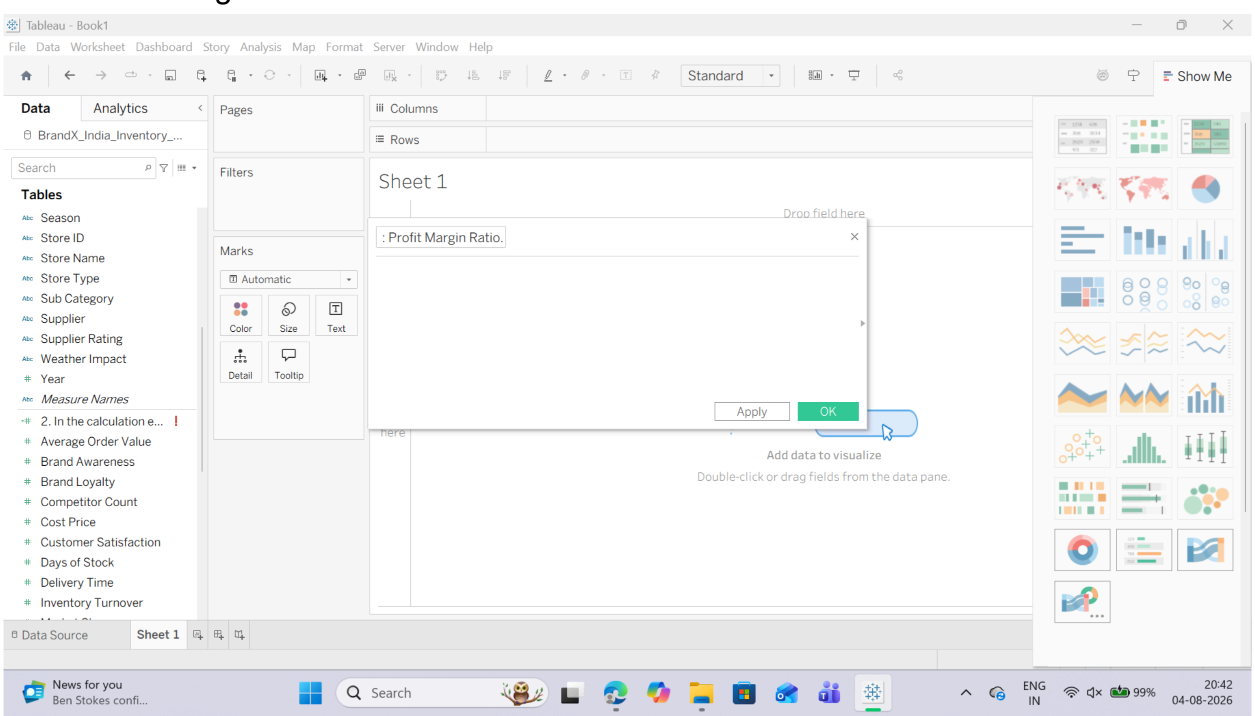
2. Building a Basic Custom Calculation

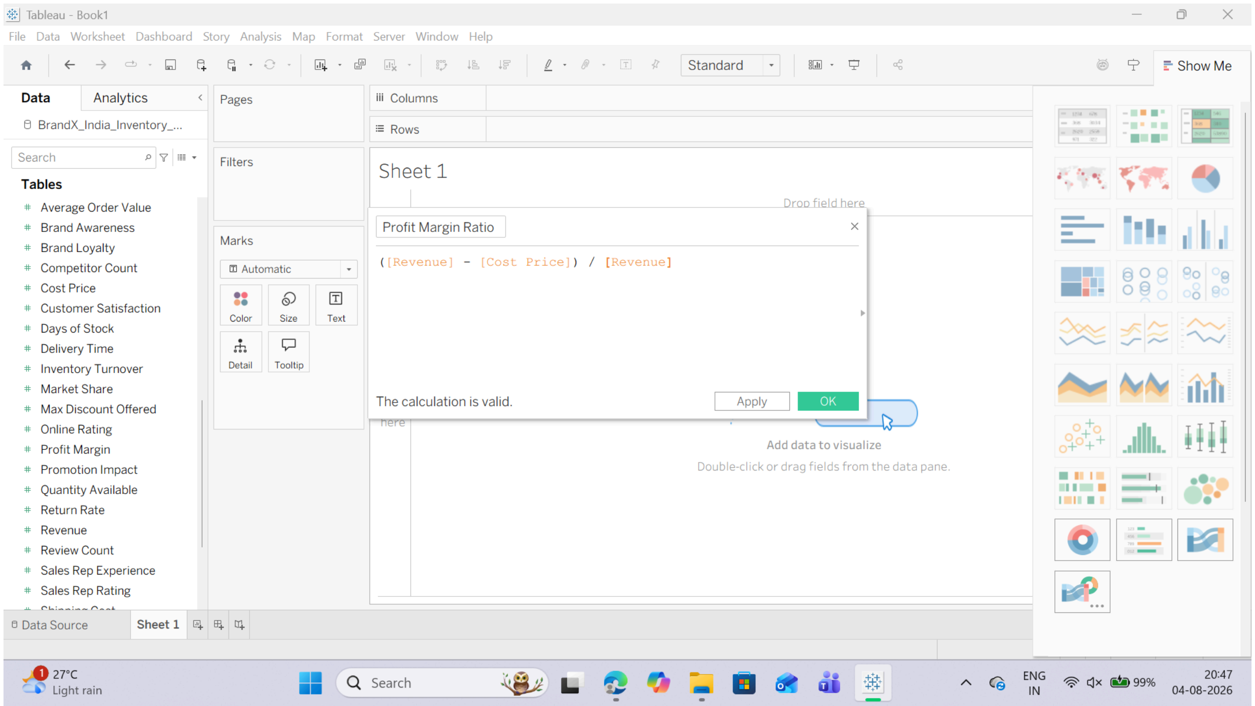
1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane (or click the small downward triangle next to the search bar) and select Create Calculated Field...



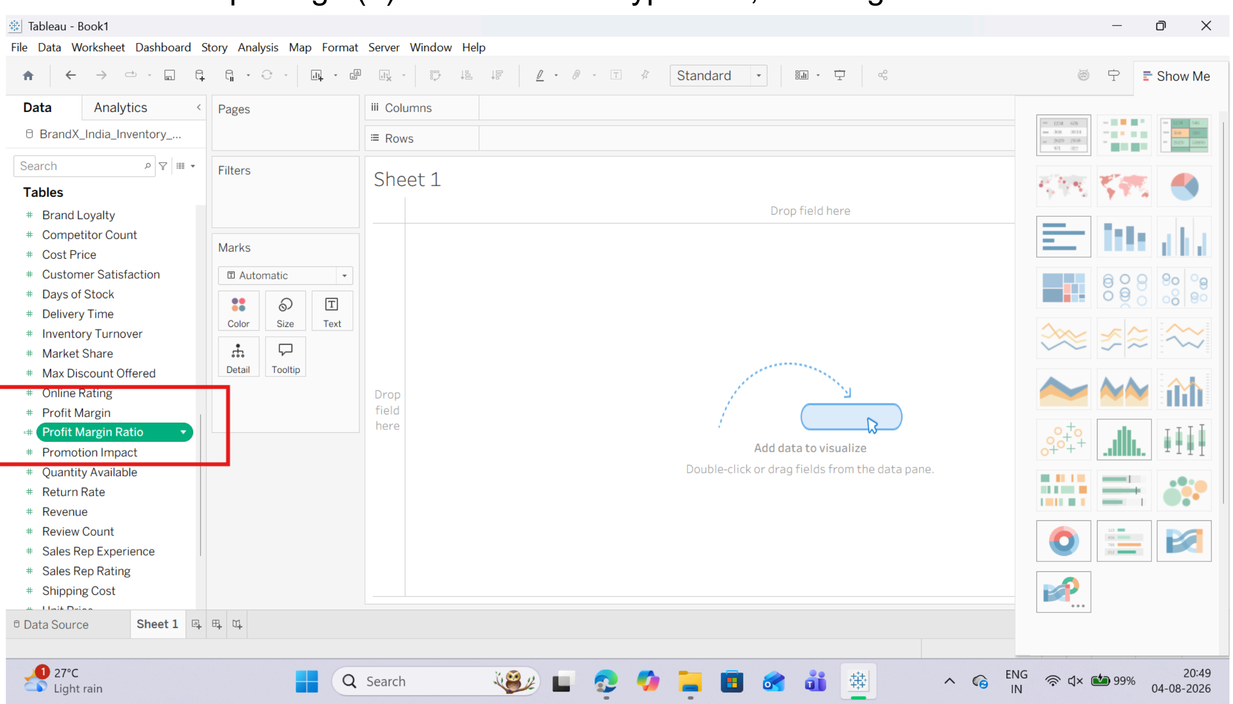
2. In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.



3. In the main workspace box, enter the following basic mathematical formula:
4. Look at the bottom-left corner of the editor box. It should say "The calculation is valid." Click OK.

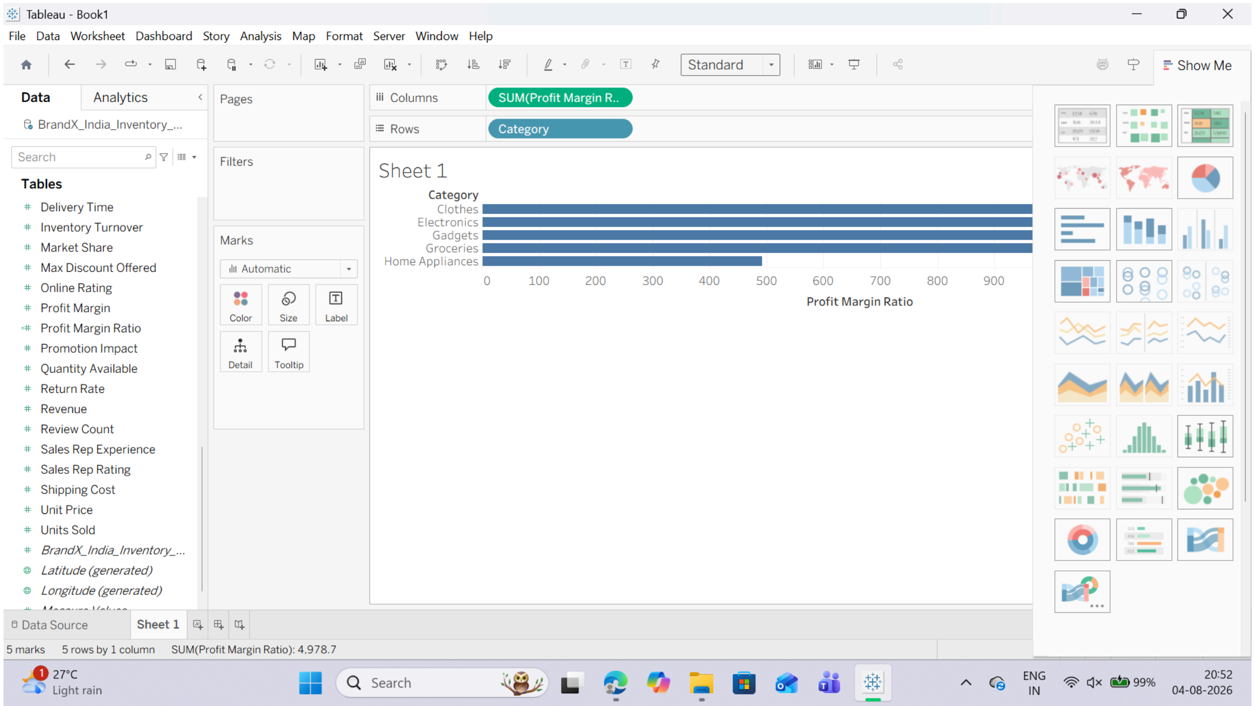


5. Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.

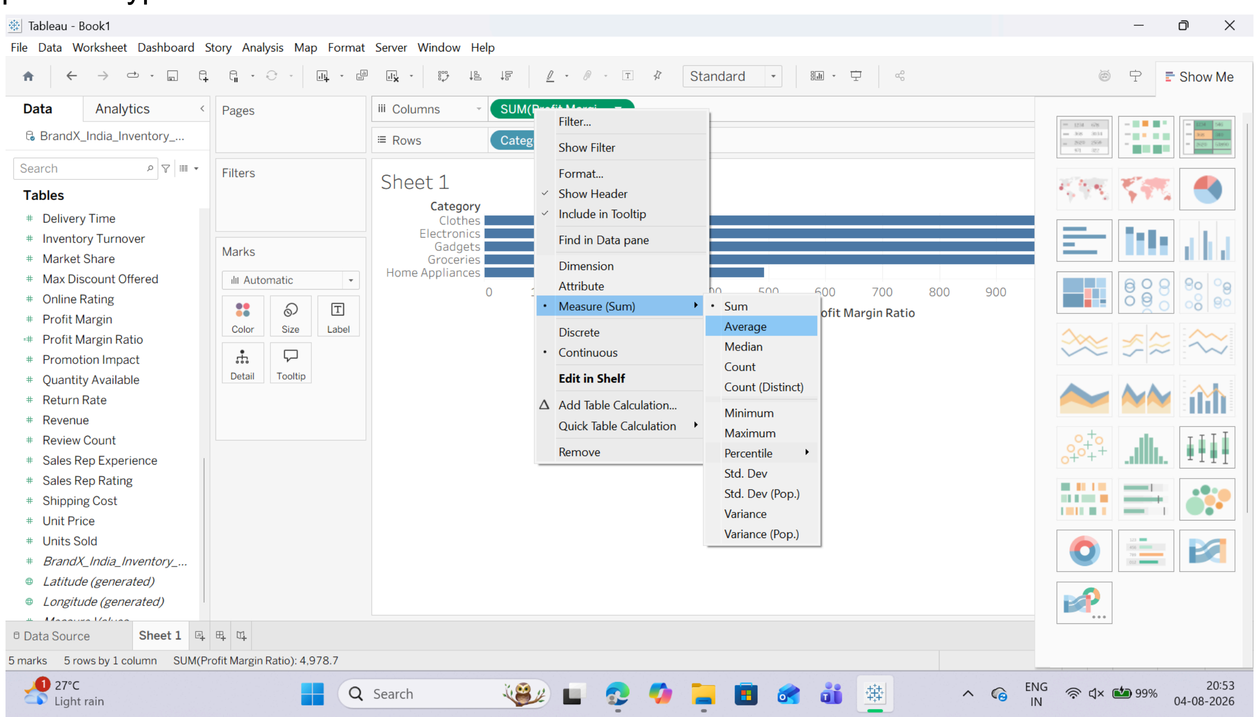


2. Putting the Field to Use

1. Drag the Category dimension from your Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.



3. Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.



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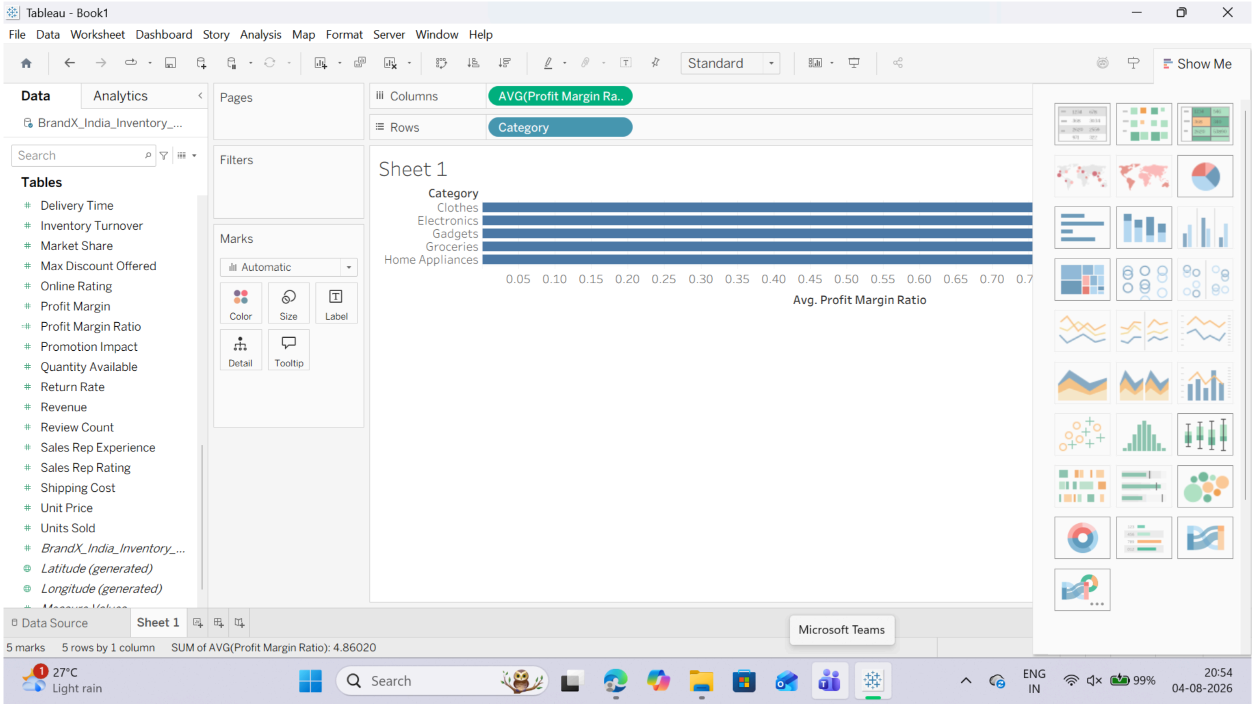
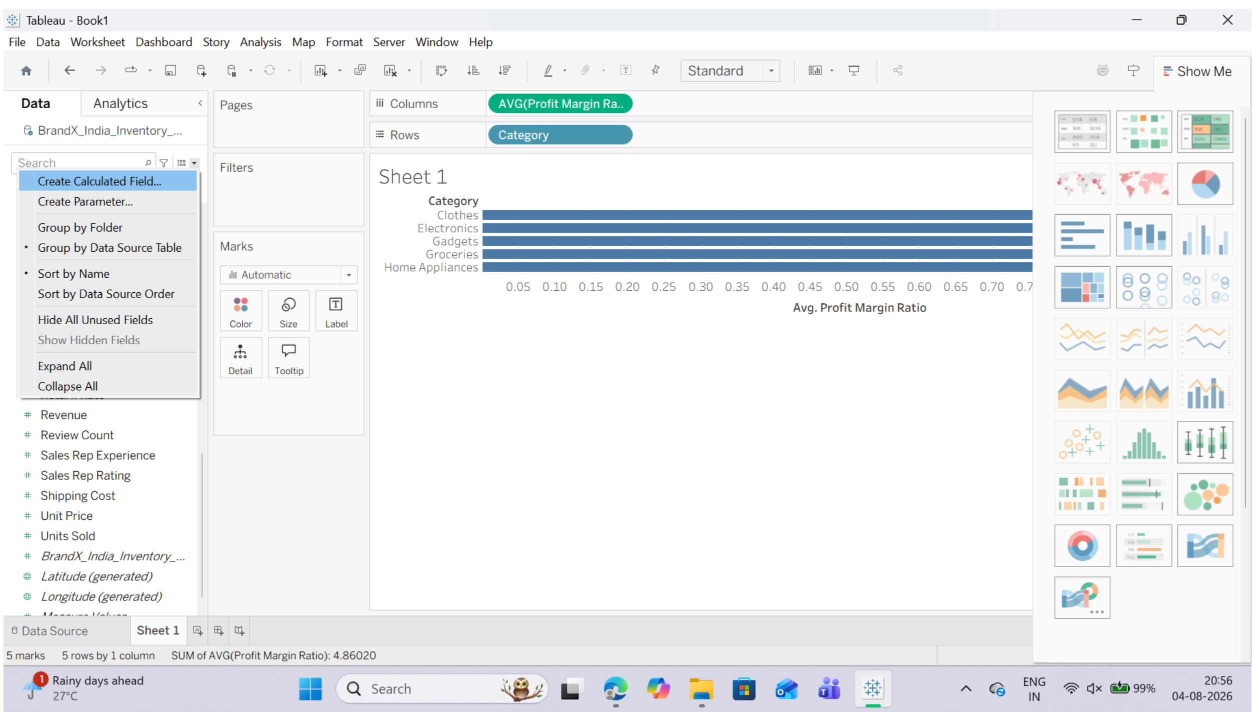


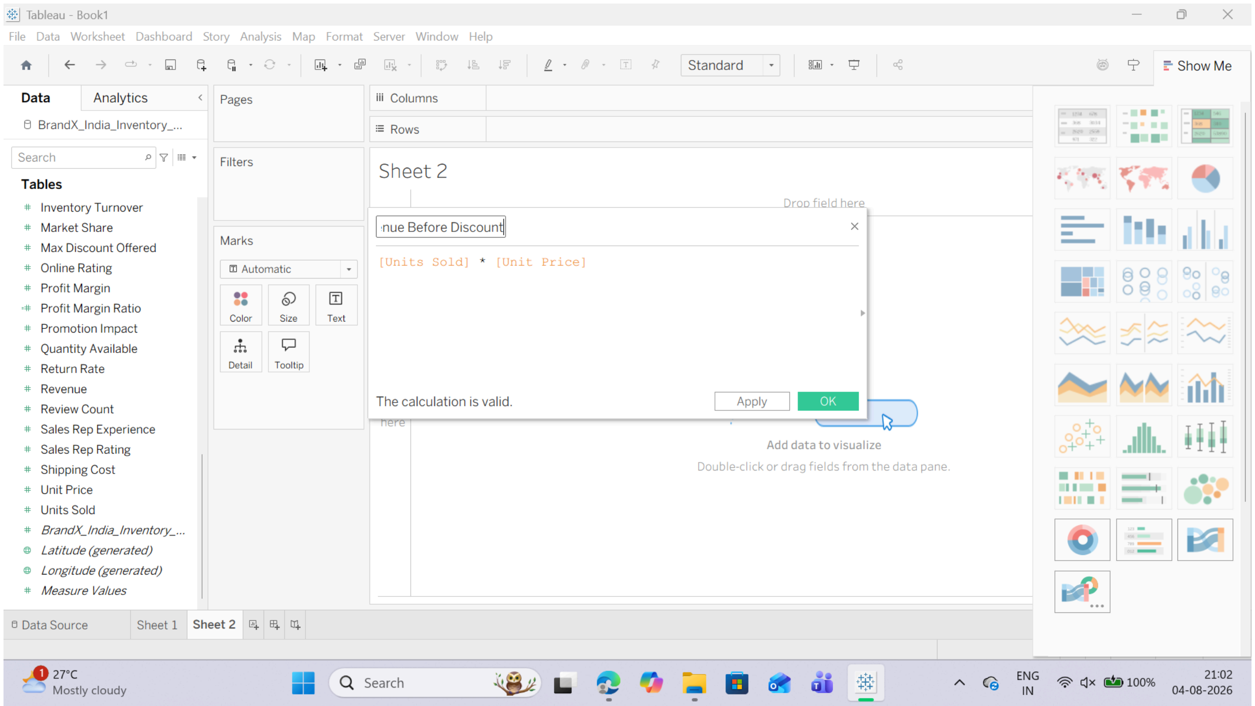
B: Performing Row-Level Calculations

1. Creating a Row-Level Calculated Field

1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...



2. Name the field: Total Revenue Before Discount.
3. In the calculation formula canvas area, input the following expression:
4. Verify the message at the bottom reads "The calculation is valid" and click OK.



2. Putting the Field to Use

1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf.
3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.

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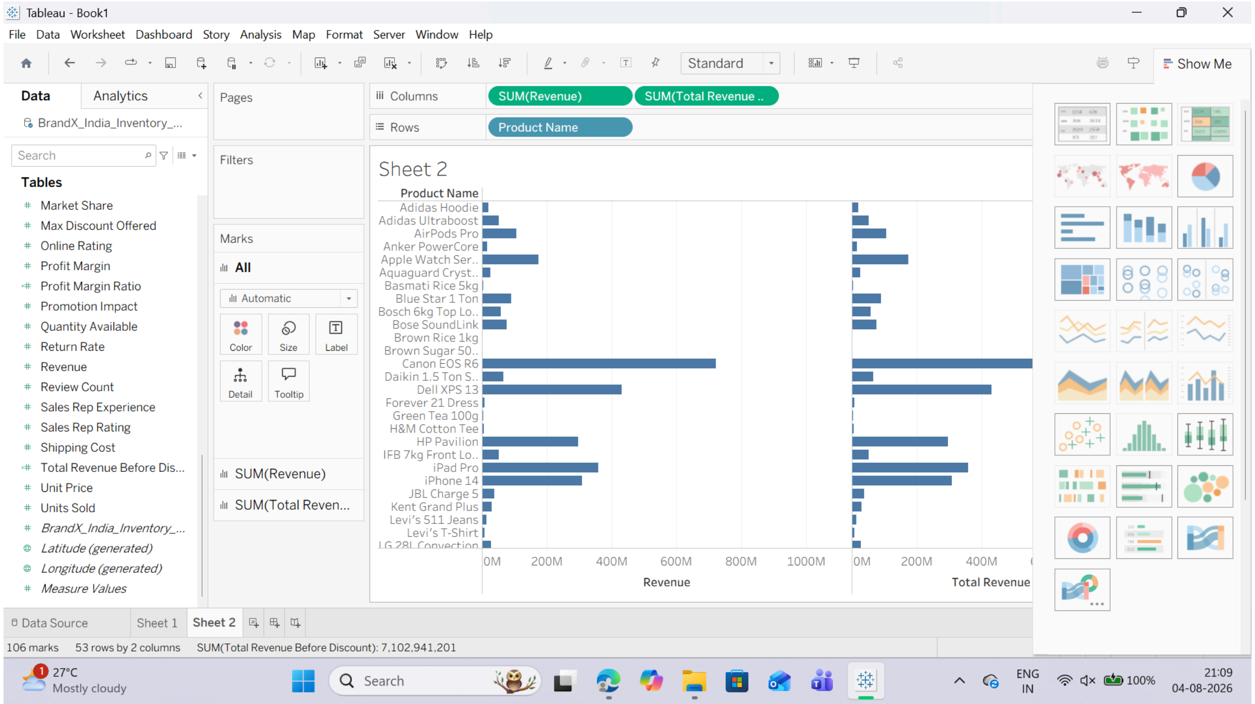
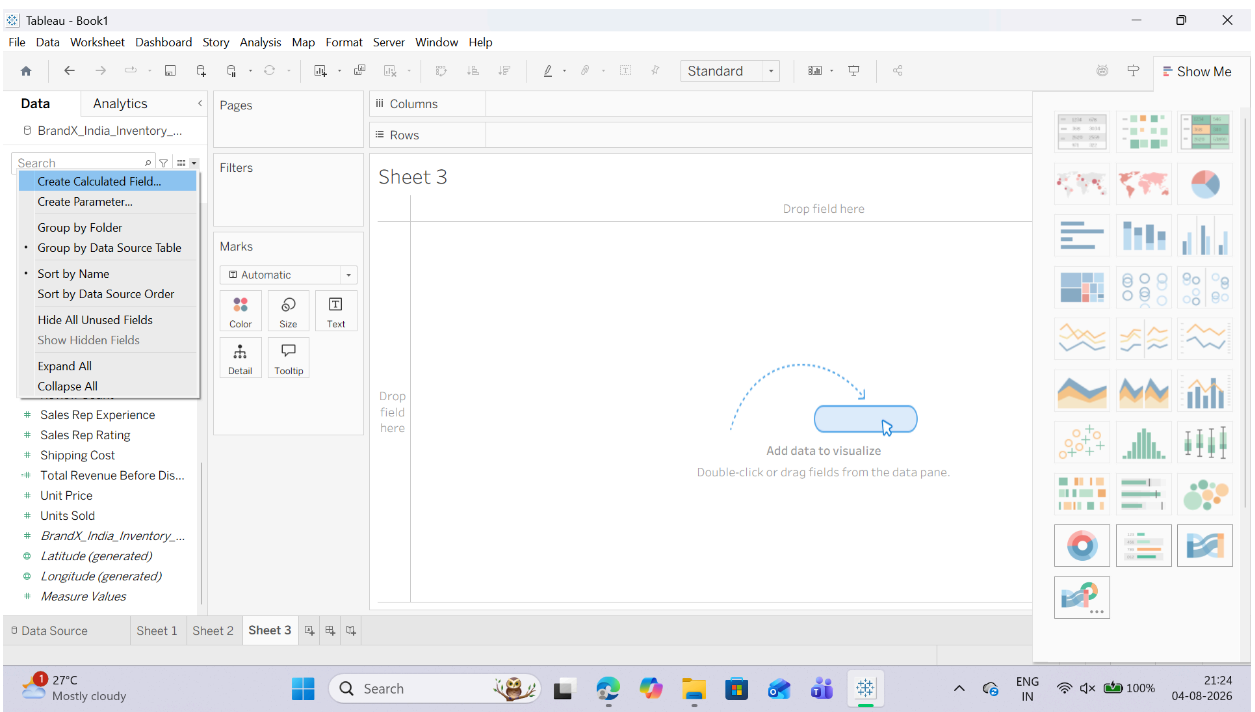


C: Performing Aggregate-Level Calculations

1. Creating an Aggregate Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field…



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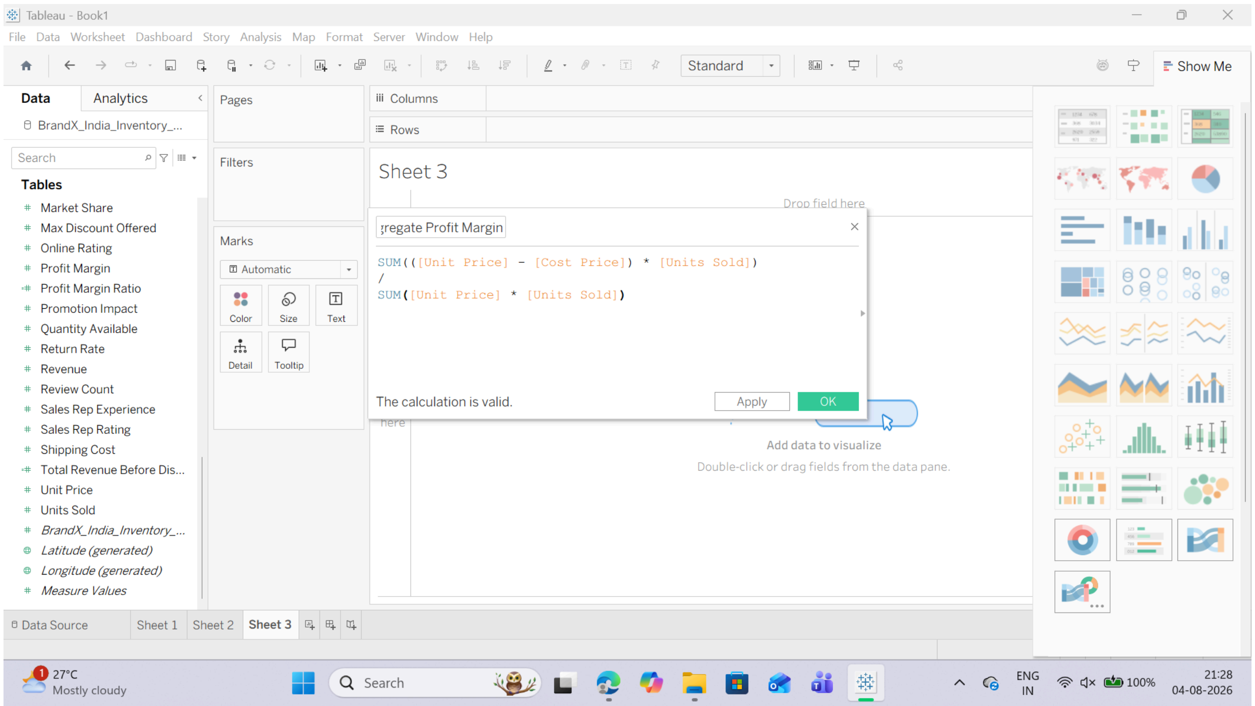
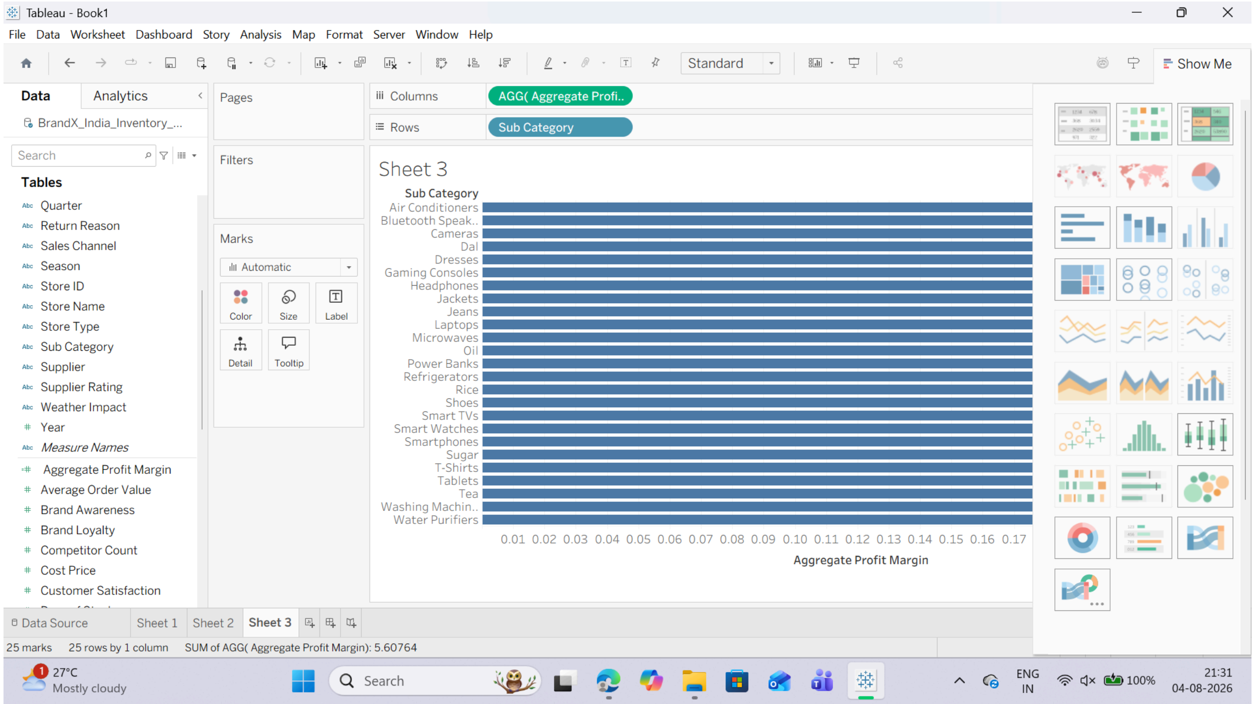
2. Name the field: Aggregate Profit Margin.
3. In the calculation formula window, type the following expression:
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



2. Putting the Field to Use

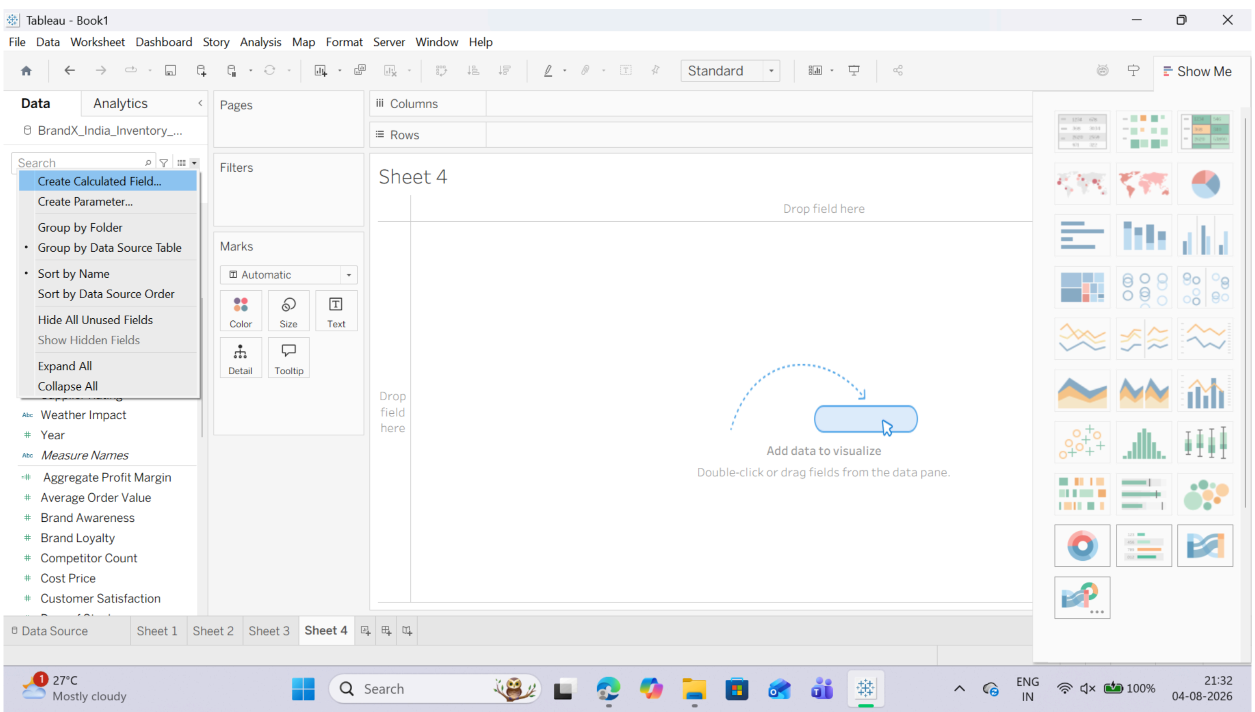
1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.
3. Observe closely: Look at the pill on the Columns shelf. It displays as AGG(Aggregate Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.



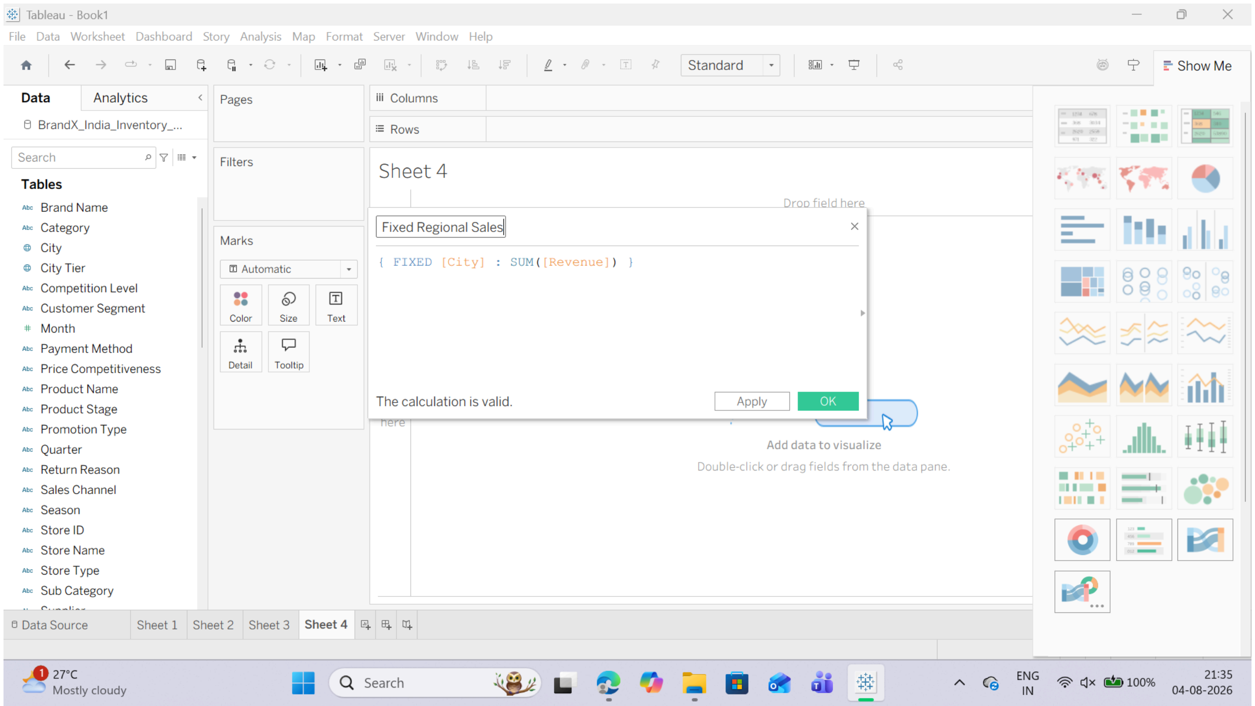
D: Implementing FIXED Level of Detail (LOD) Calculations

1. Creating a FIXED LOD Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field…



2. Name the field: Fixed Regional Sales.
3. In the calculation formula window, type the following expression using curly braces {}:
4. Look at the bottom left to ensure it reads "The calculation is valid" and click OK.



2. Putting the Field to Use

1. Drag the City dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to City.
3. Drag the revenue measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.

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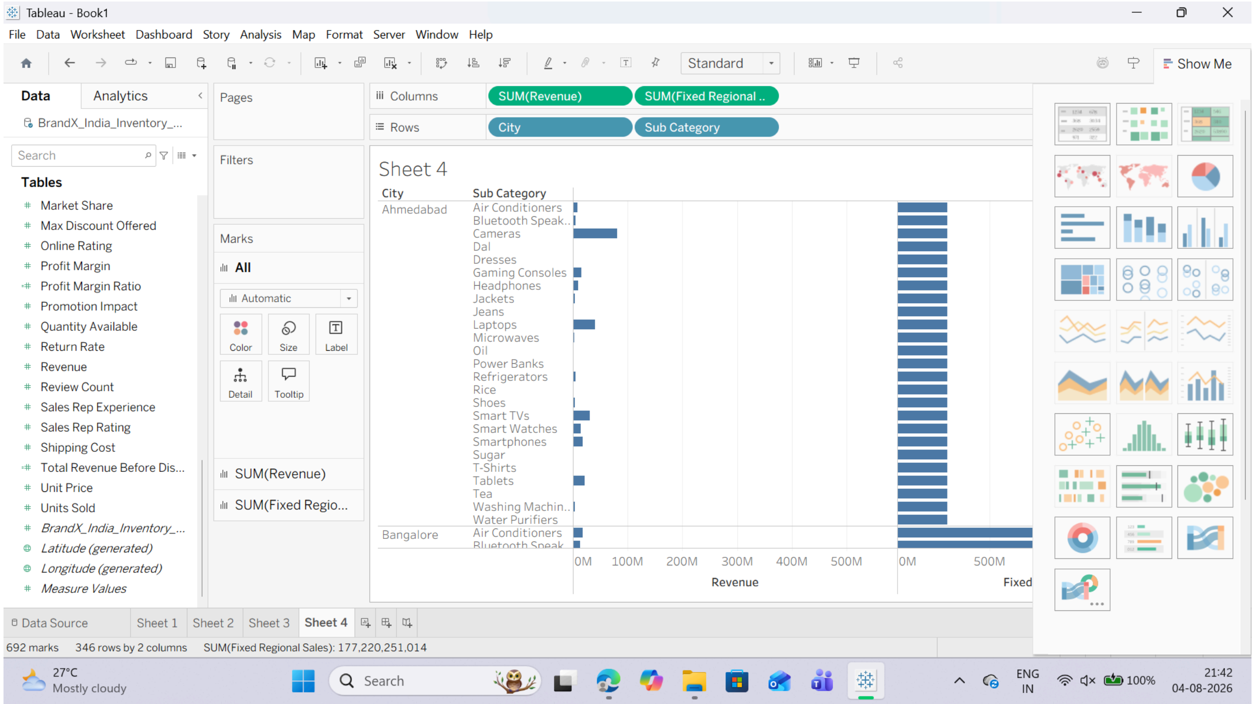
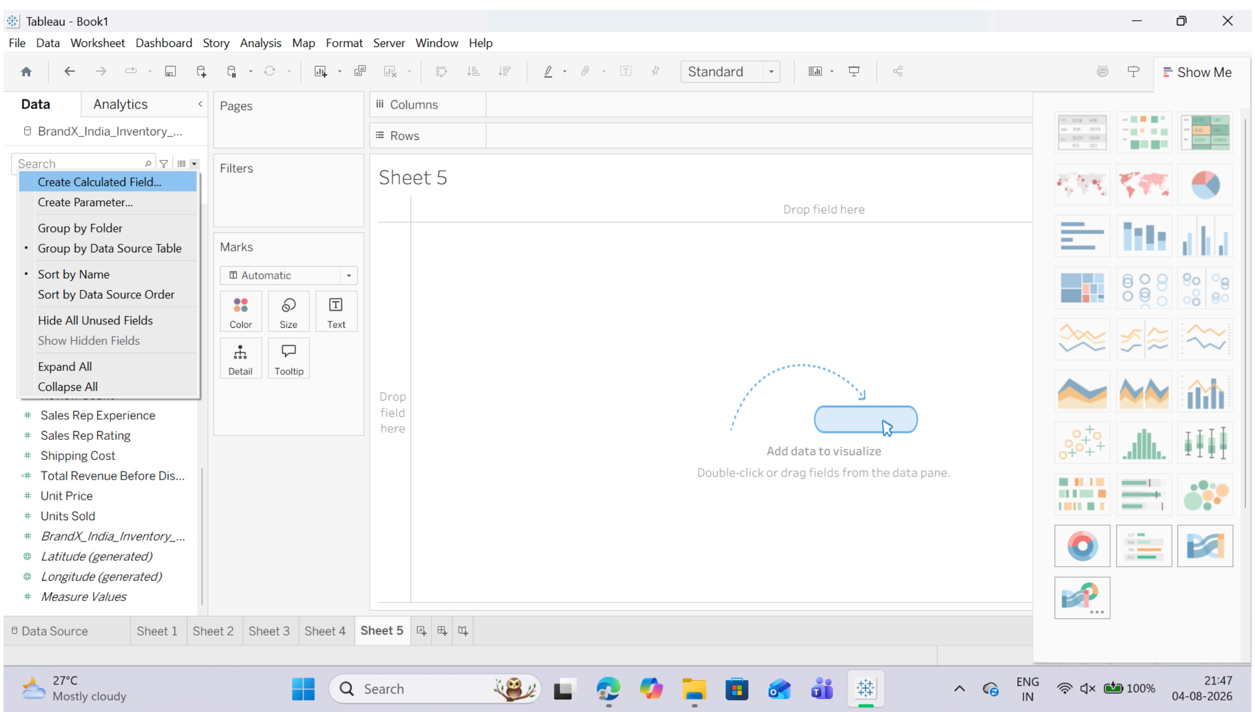


E: Using INCLUDE and EXCLUDE LOD Expressions

1. Creating an INCLUDE LOD Calculated Field

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...



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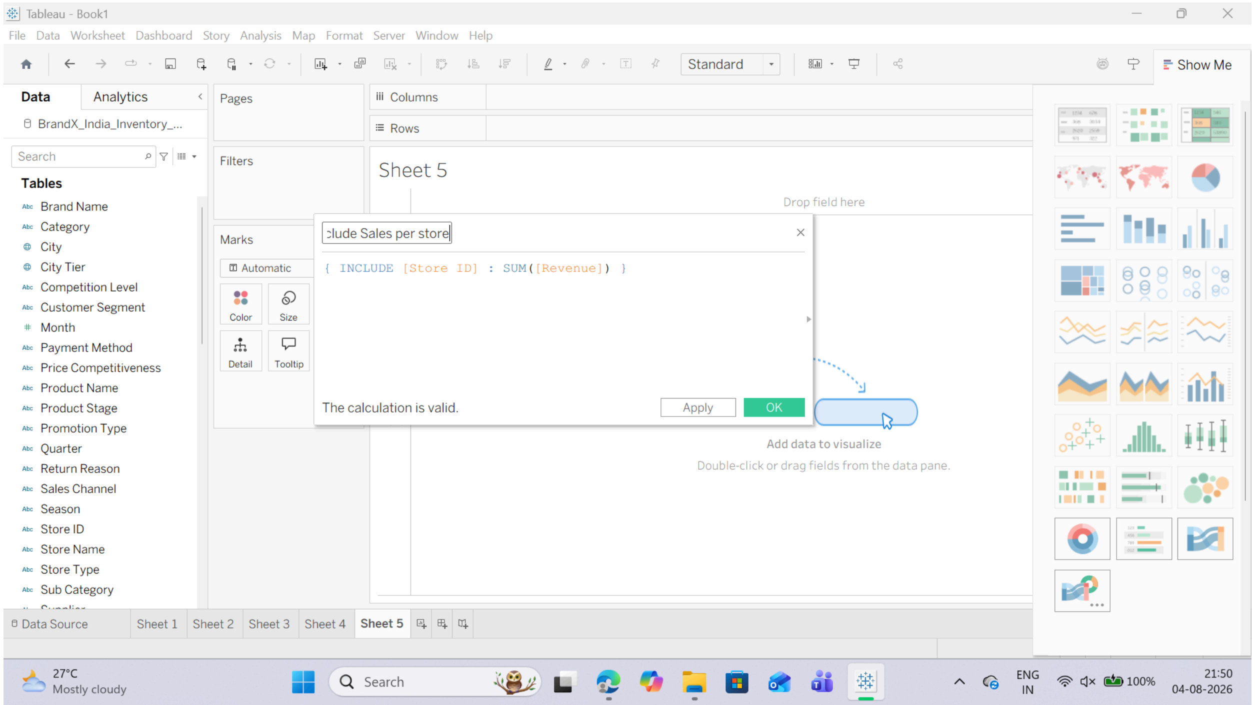
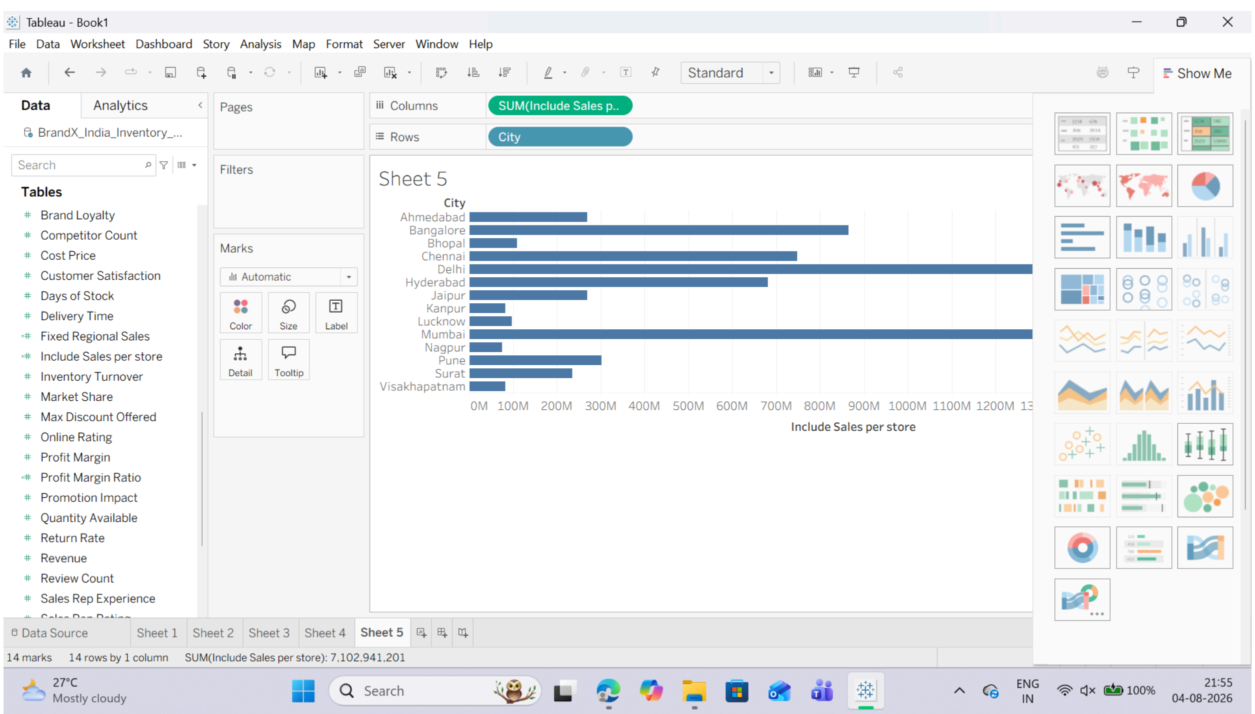
2. Name the field: Include Sales per Store.
3. In the calculation formula canvas area, input the following expression:



2. Putting the INCLUDE Field to Use

1. Drag the City dimension from the Data Pane and drop it onto the Rows shelf.

2. Drag your newly created Include Sales per Store calculated field and drop it onto the Columns shelf.



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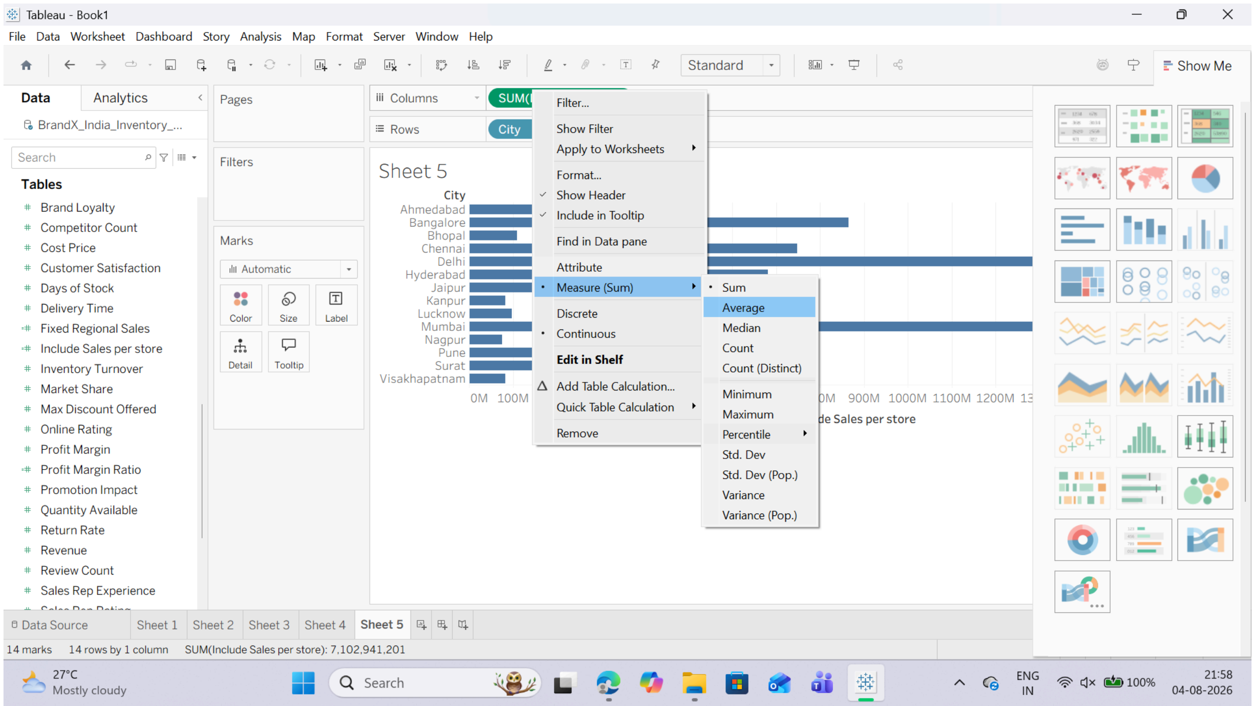
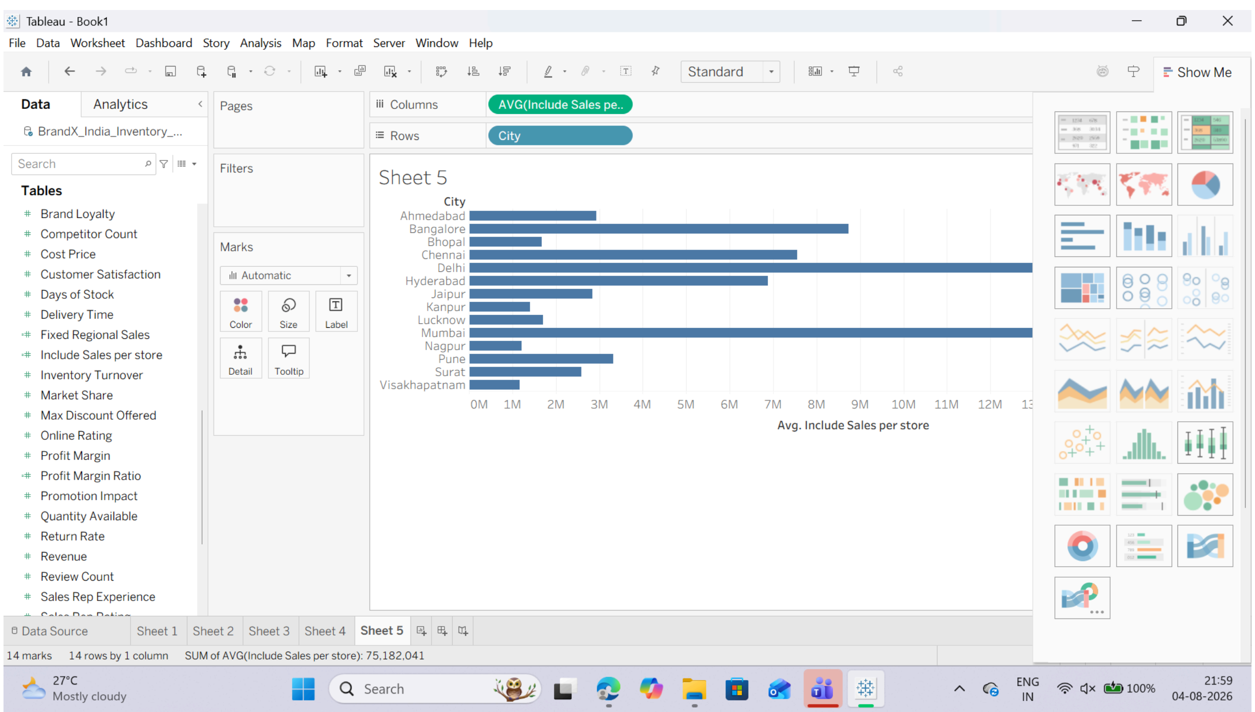
3. Right-click the Include Sales per Store pill on the Columns shelf, hover over Measure (Sum), and change it to Average.





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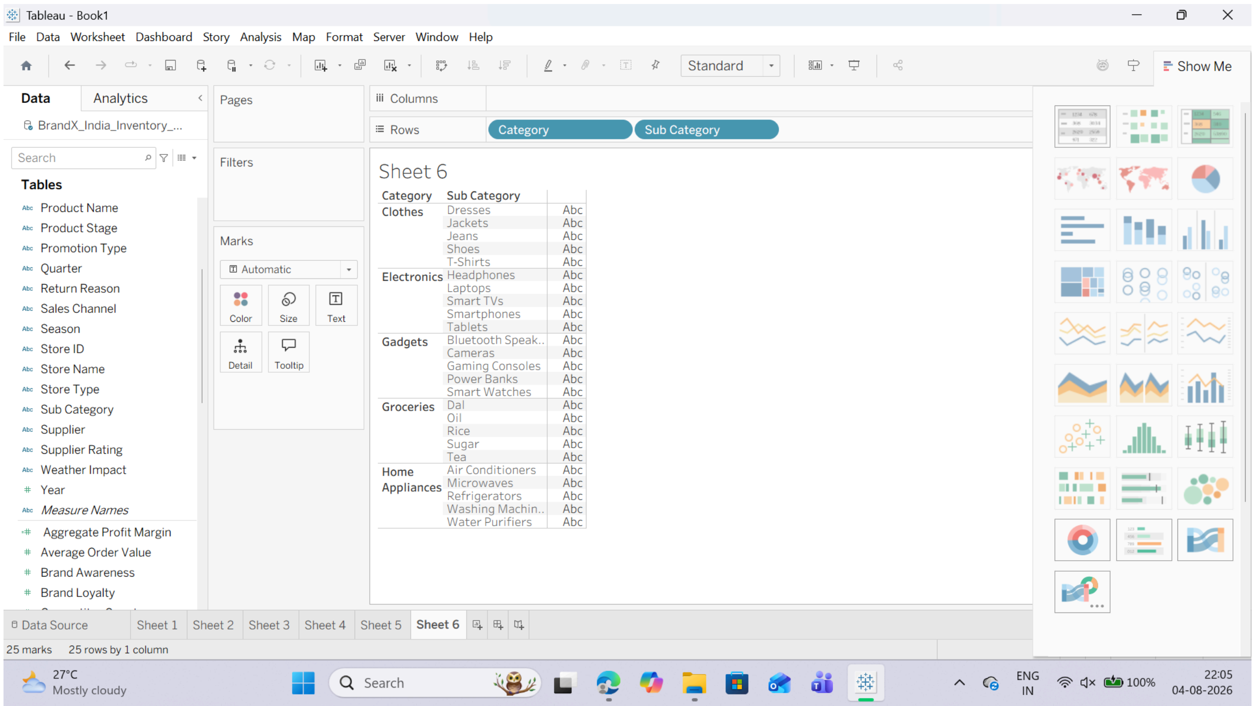
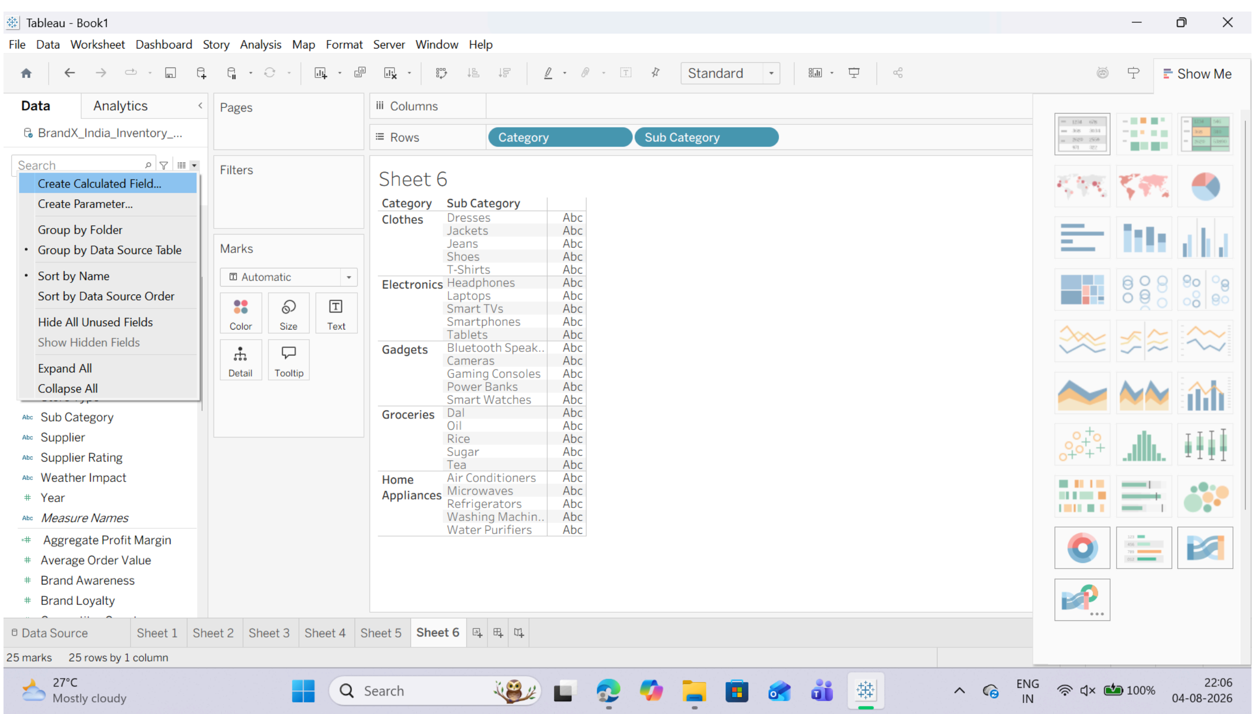
3. Creating an EXCLUDE LOD Calculated Field

1. Open a new blank worksheet.
2. Drag Category onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).



4. Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



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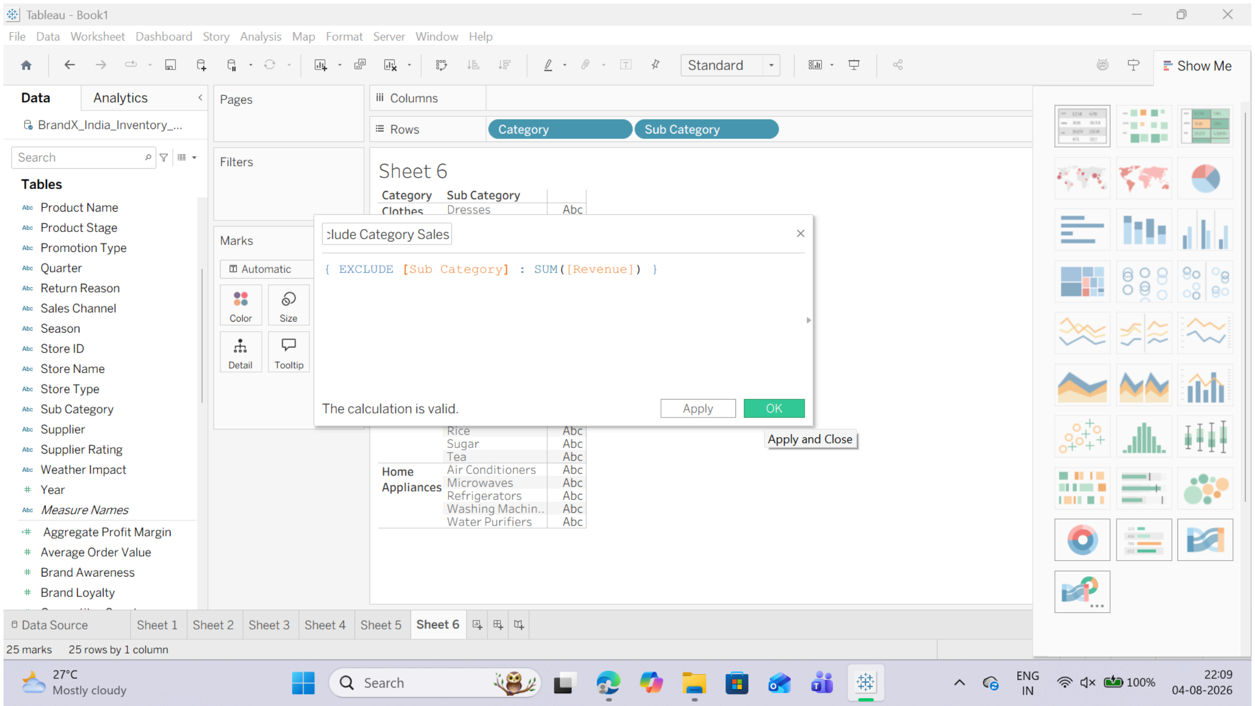
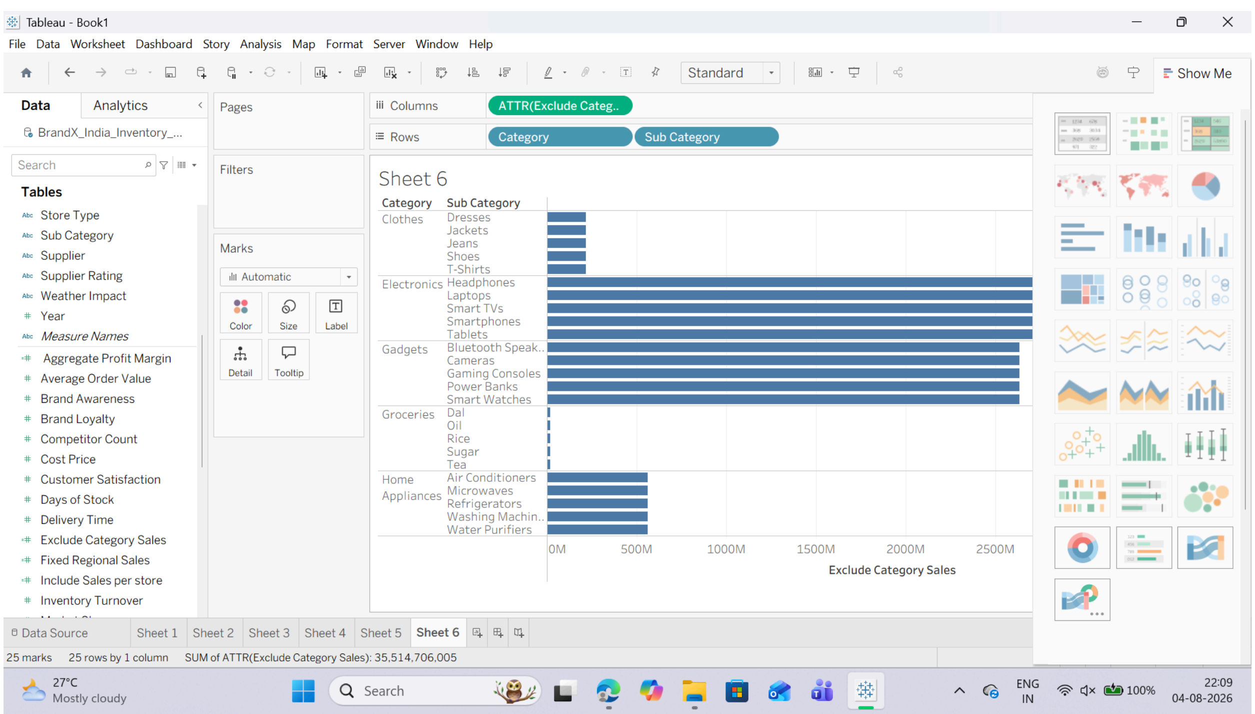
5. Name the field: Exclude Category Sales.
6. Input the following expression:
7. Click OK and drag this new field onto the Columns shelf.



7. Click OK and drag this new field onto the Columns shelf.



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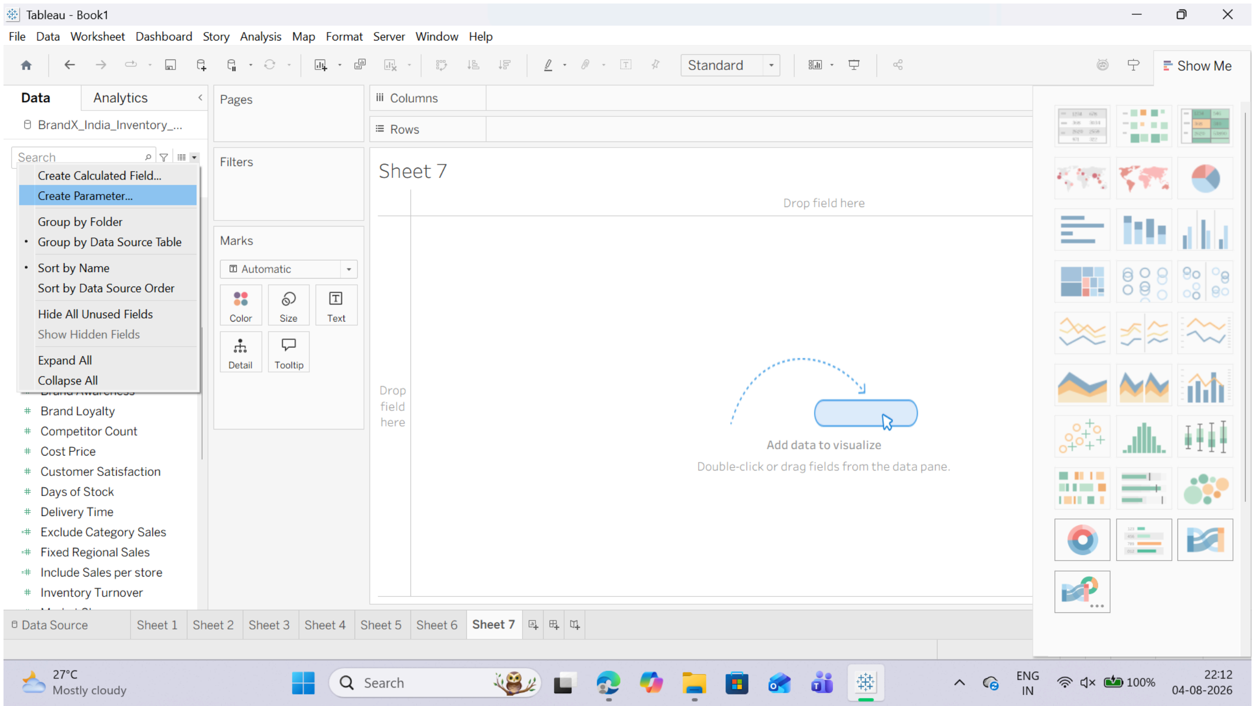
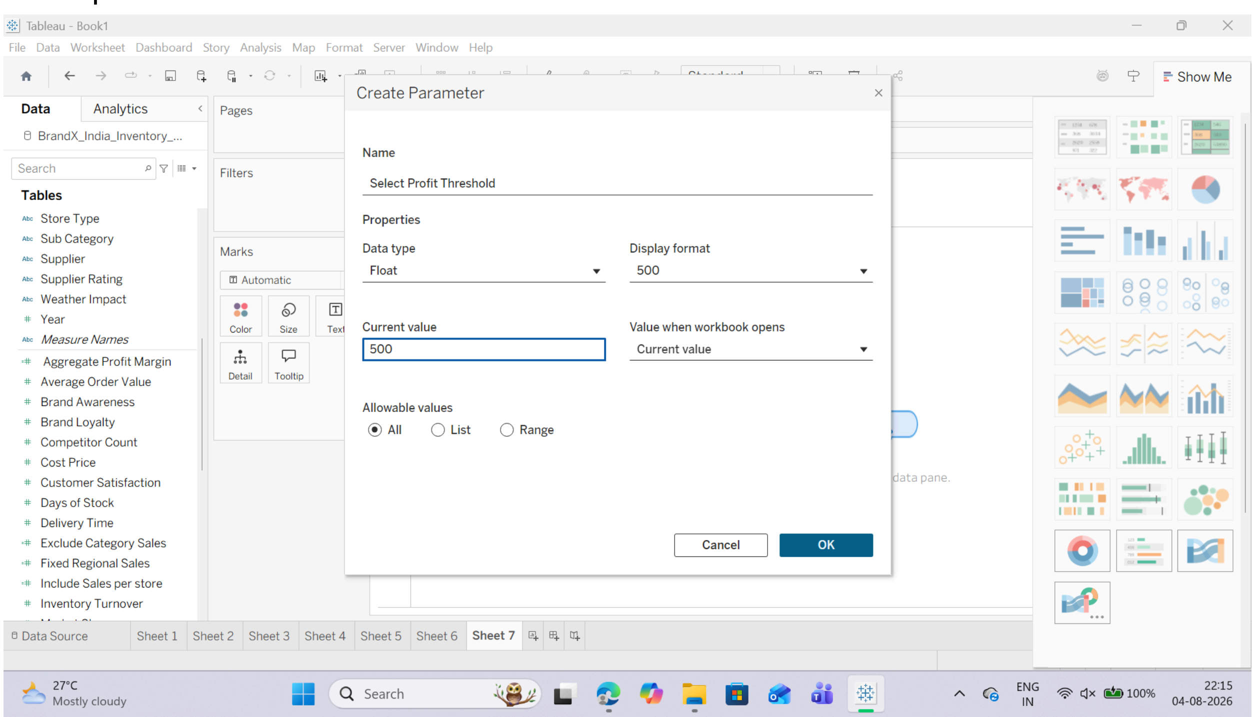
F: Creating and Using Parameters for Dynamic Analysis

1. Creating a Parameter

1.Click the small drop-down arrow next to search bar at top and choose Create Parameter



2. Name the parameter: Select Profit Threshold.
3. Set the Data Type to Float (or Integer).
4. Set the Current Value to 500.
5. Keep the Allowable values set to All and click OK.



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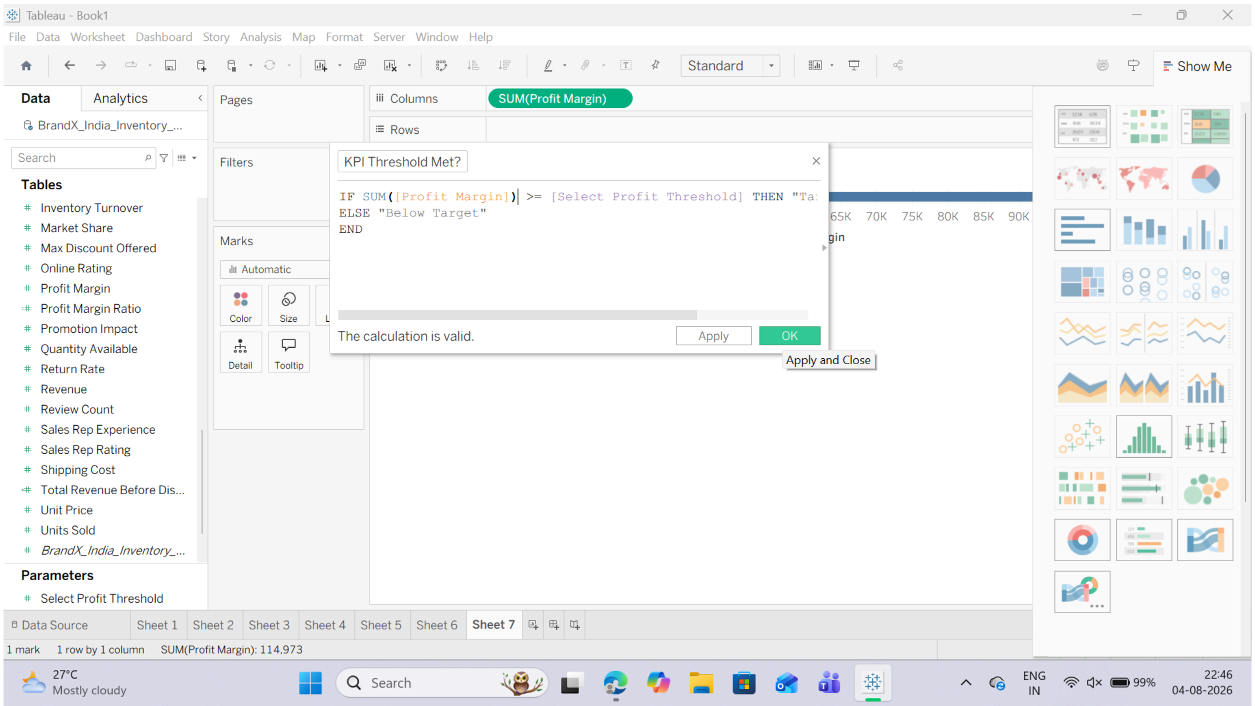
2. Linking the Parameter to a Calculated Field

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle next to search bar and select Create Calculated Field...
2. Name this field: KPI Threshold Met?.
3. Input the following logical IF-THEN conditional statement:



3. Putting the Interactive Parameter to Use

1. Open a new blank worksheet.
2. Drag the City dimension from your Data Pane and drop it onto the Rows shelf.
3. Drag the Revenue measure and drop it onto the Columns shelf.
4. Drag your newly created calculated field KPI Threshold Met? and drop it directly onto the Color tile in the Marks card.

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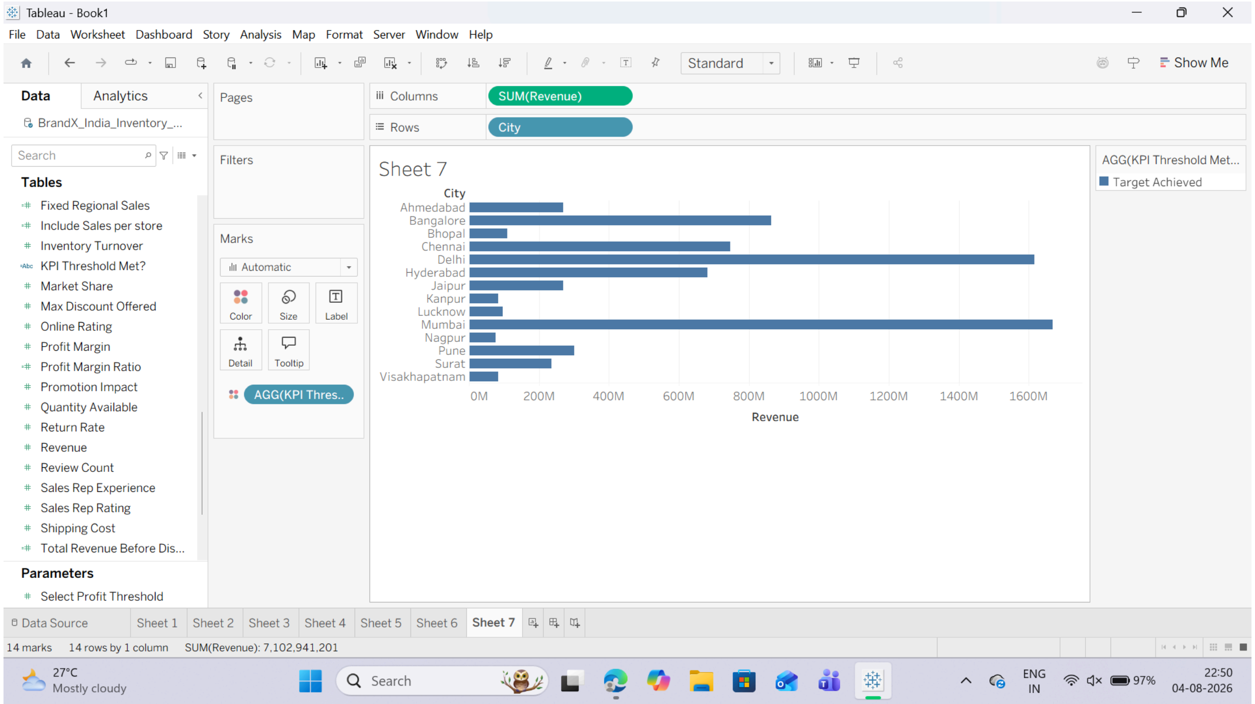
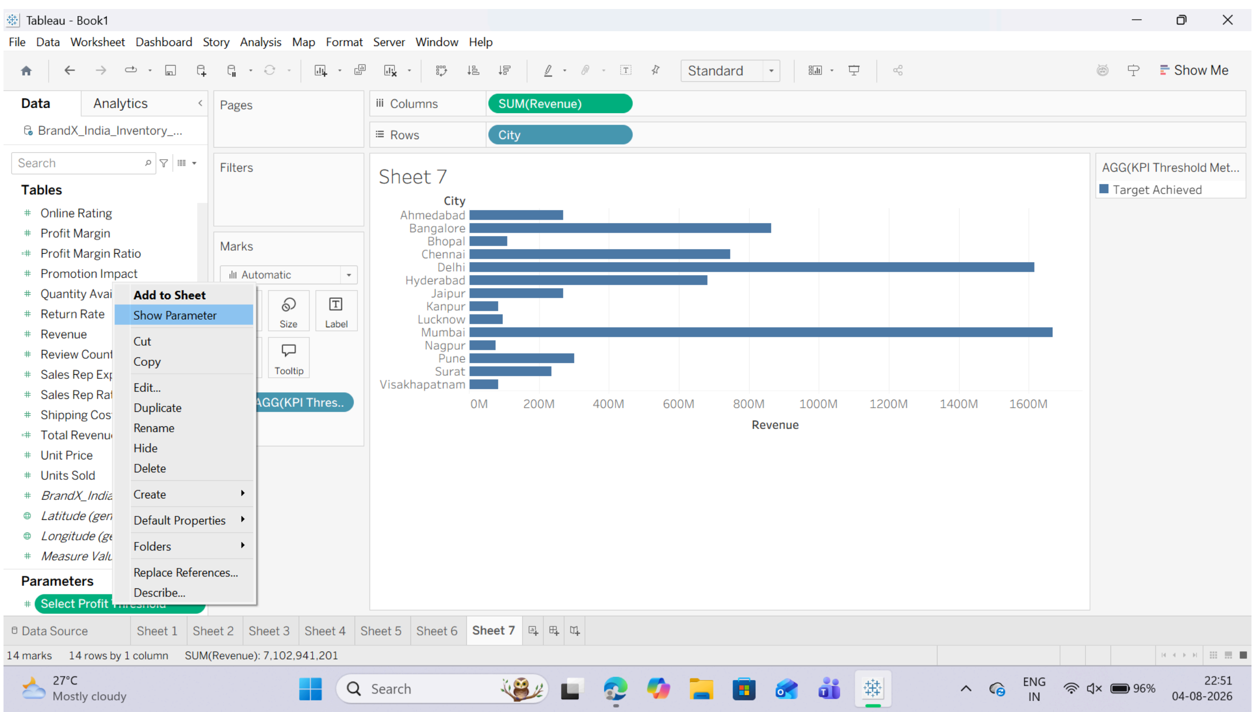


5. Show the Parameter Control: Go down to the very bottom of the Data Pane under the Parameters header section. Right-click on your Select Profit Threshold parameter block and choose Show Parameter Control.



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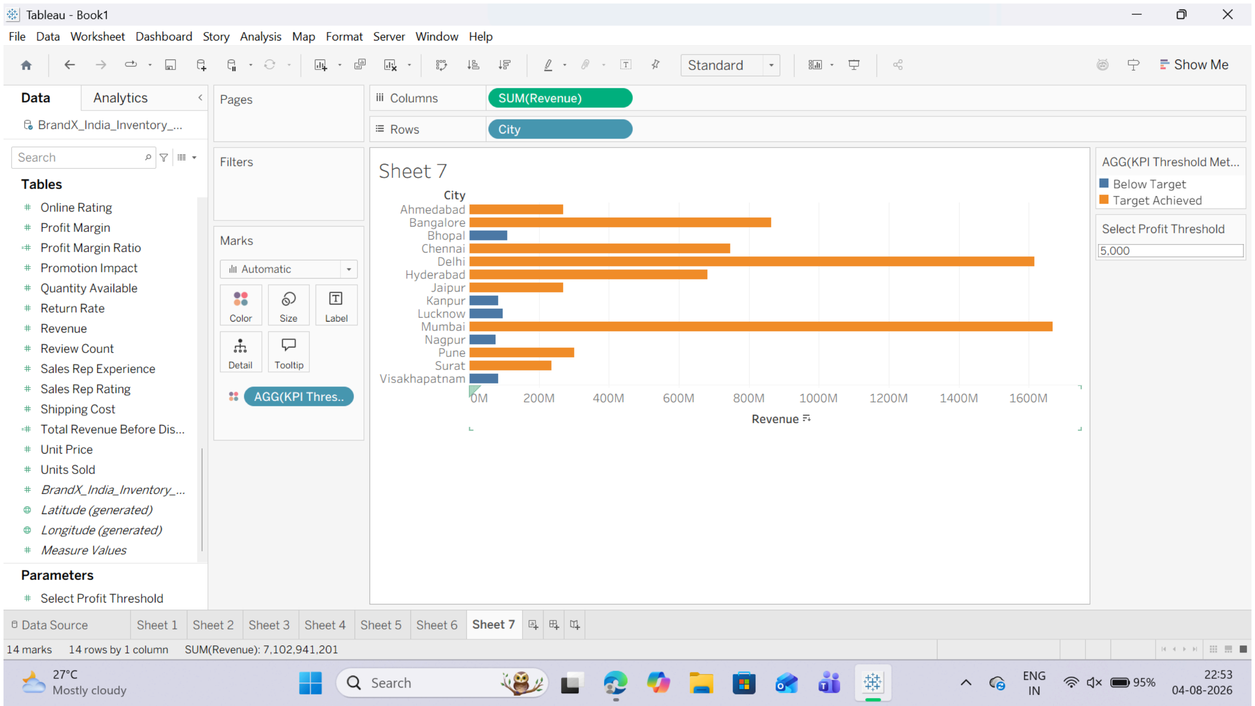
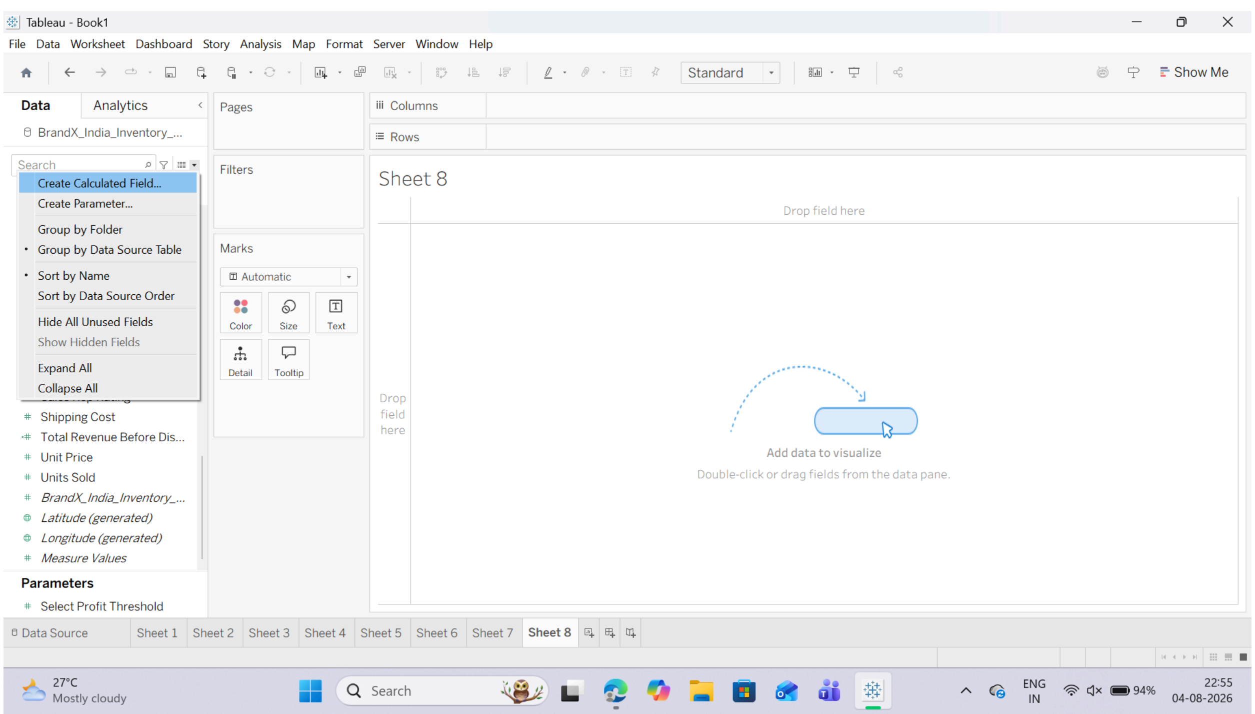


G: Apply Calculations to Fix Data Issues and Enhance Insights

1. Handling Null or Missing Values with IFNULL

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...



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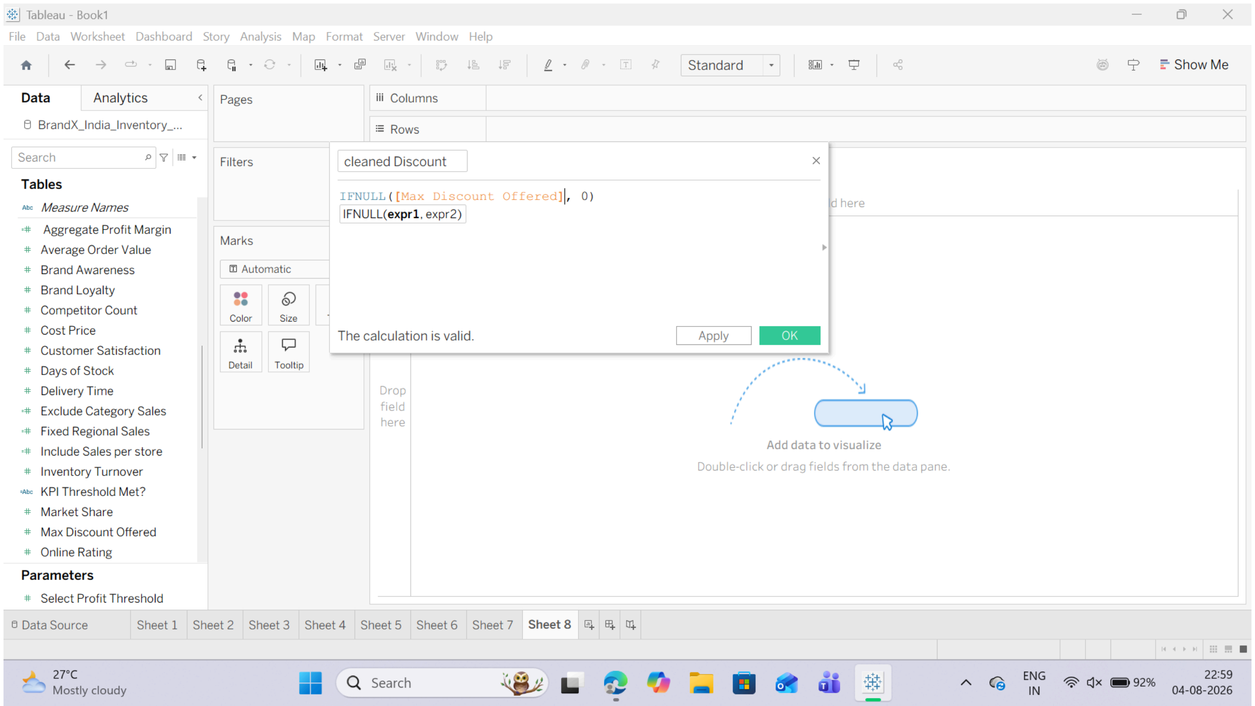
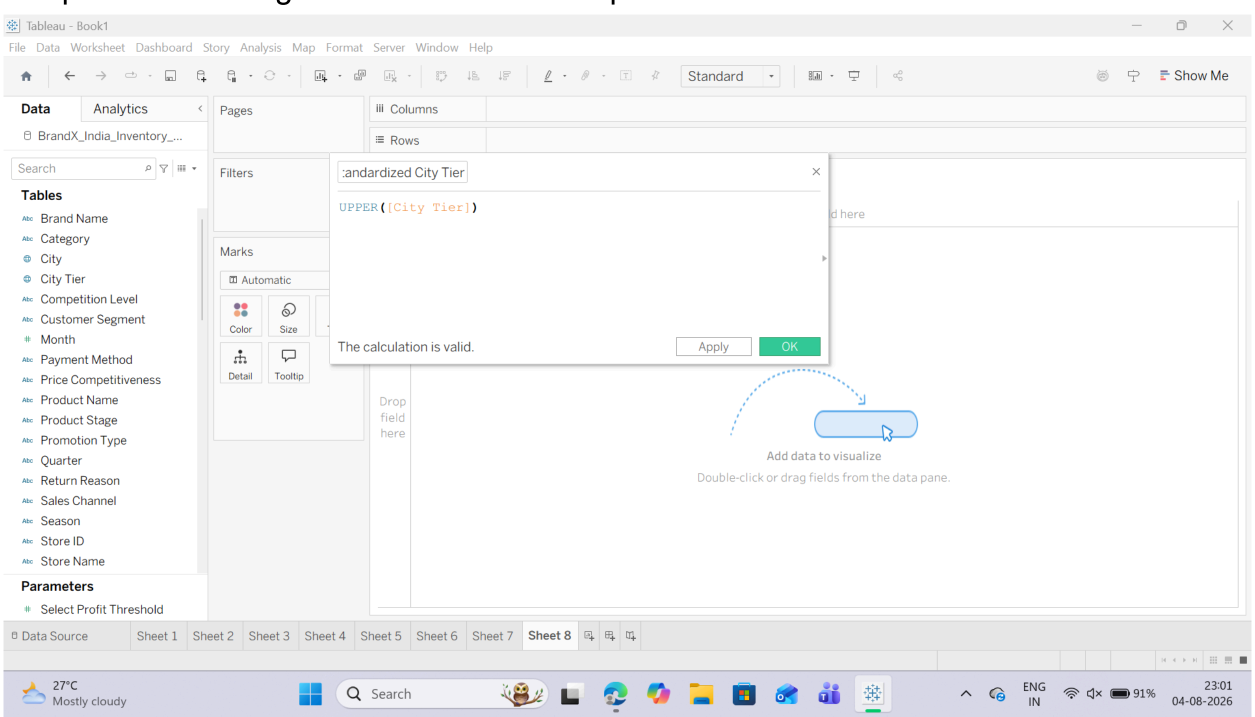
2. Name the field: Cleaned Discount.
3. Input the following expression:



2. Segmenting Data for Insights using String Functions (UPPER/LOWER)

1. Create a new Calculated Field named Standardized City Tier.
2. Input the following text transformation expression:



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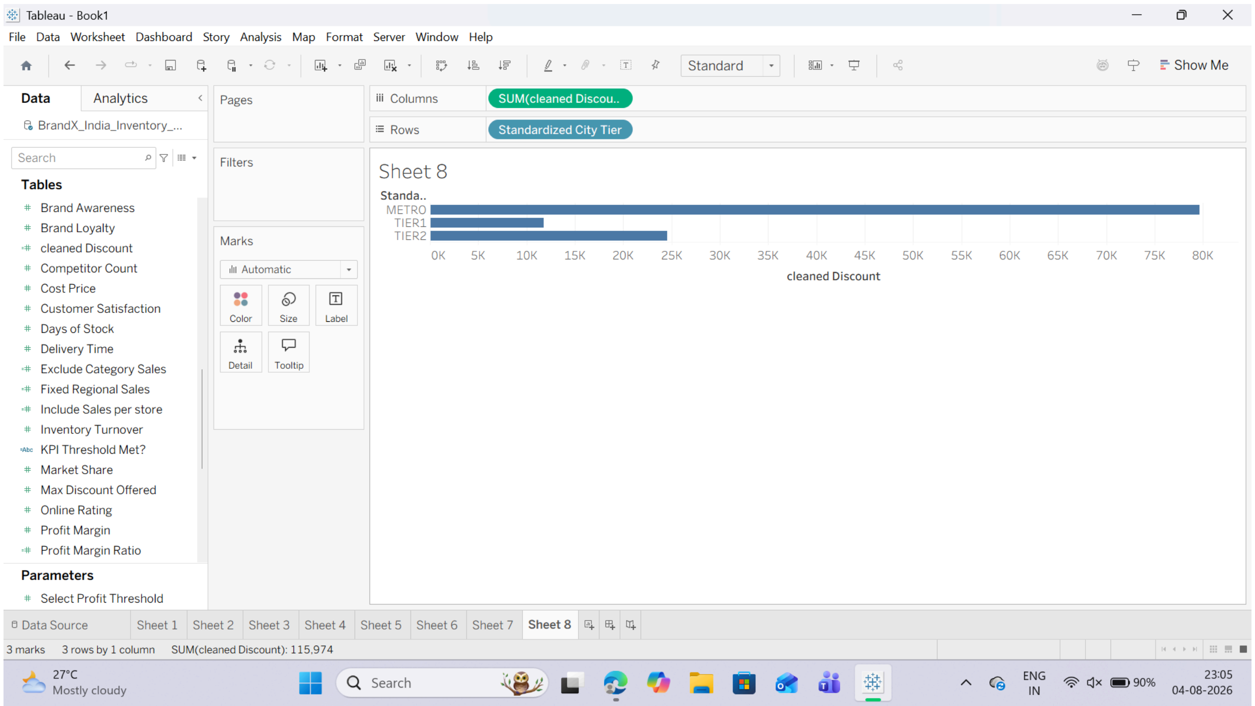
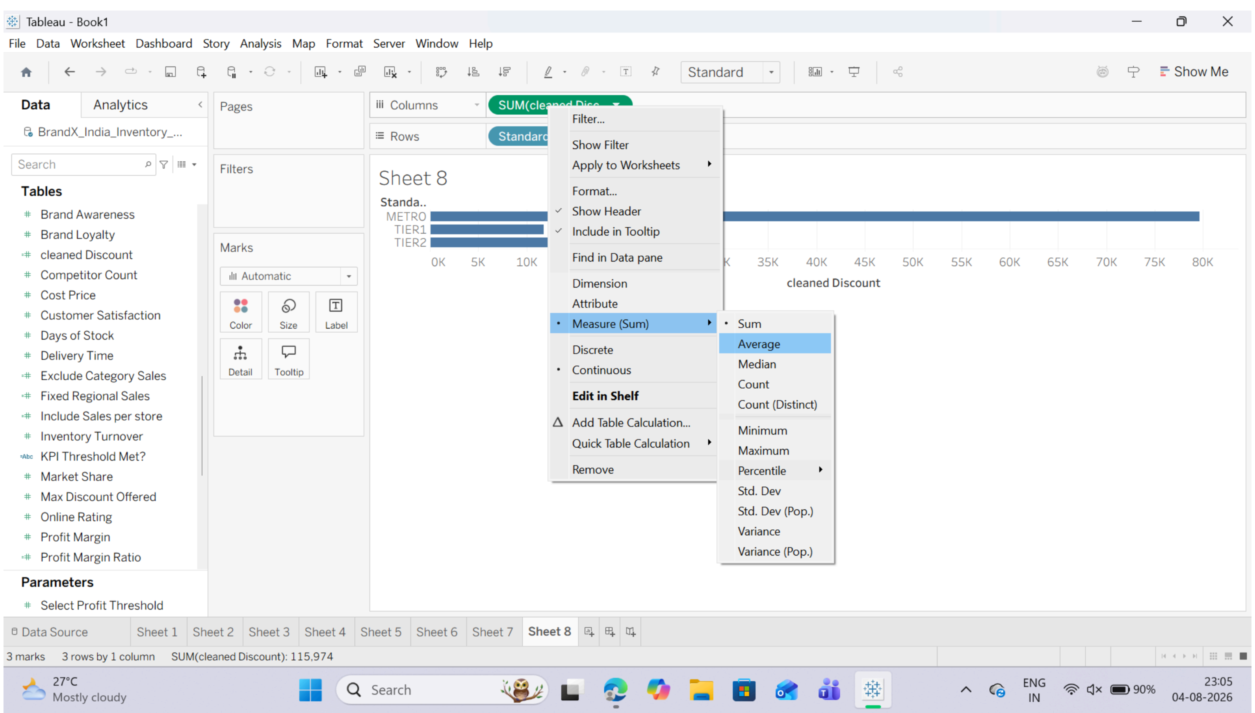
3. Putting the Cleaned Fields to Use

1. Open a new blank worksheet.
2. Drag your newly created Standardized City Type dimension from the Data Pane and drop it onto the Rows shelf.
3. Drag the Cleaned Discount calculated field onto the Columns shelf. Change its aggregation from Sum to Average by right-clicking the pill.





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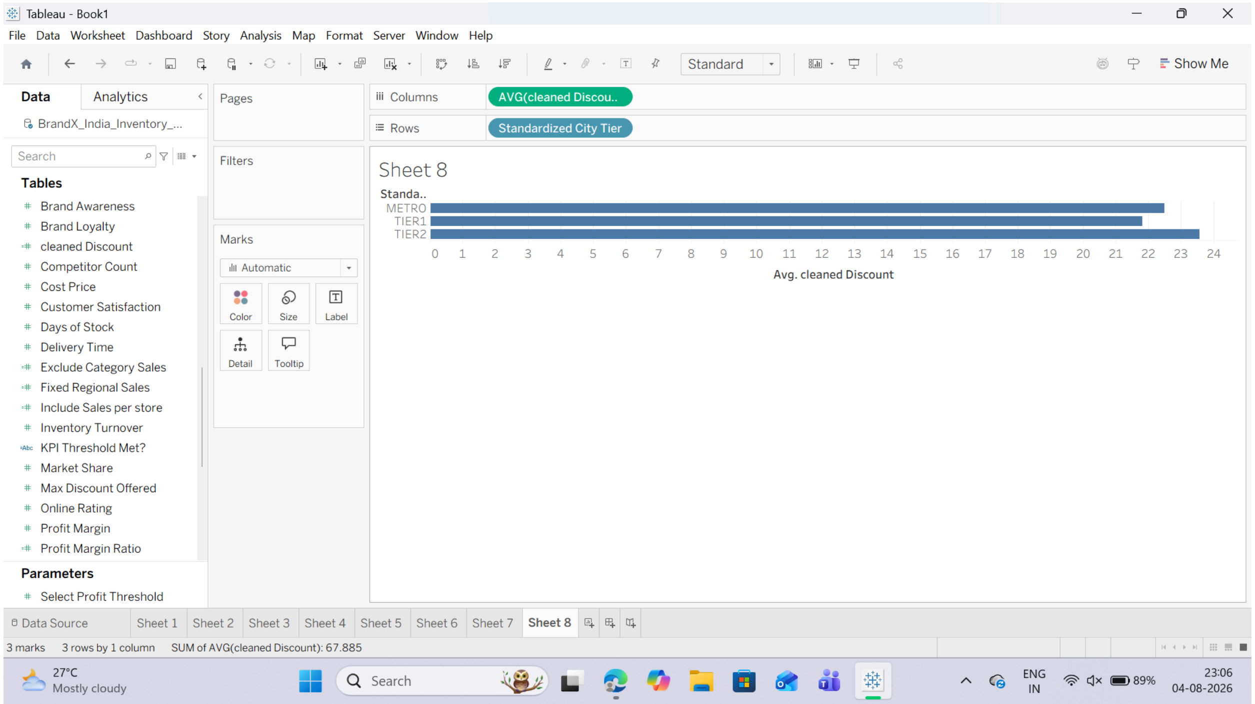

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